

OSSP 8

Question:

1. Develop a program using malloc(), calloc(), realloc(), and free(). Monitor memory allocation behavior and identify memory leaks using Valgrind.

Develop a program that demonstrates Copy-on-Write behavior after a fork(). Analyze memory usage before and after modifying data in the child process.

Solution:

```
ubuntu@ubuntu:~$ nano memory_management.c
ubuntu@ubuntu:~$
ubuntu@ubuntu:~$ gcc memory_management.c -o memory_management
ubuntu@ubuntu:~$ ./memory_management
malloc() Address: 0xae1c5e58d010
calloc() Address: 0xae1c5e58d440
realloc() Address: 0xae1c5e58d460
Memory Freed Successfully
ubuntu@ubuntu:~$
```



```
GNU nano 8.7.1 memory_management.c
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int n = 5;

    int *arr1 = (int *)malloc(n * sizeof(int));

    printf("malloc() Address: %p\n", arr1);

    for(int i = 0; i < n; i++)
        arr1[i] = i + 1;

    int *arr2 = (int *)calloc(n, sizeof(int));

    printf("calloc() Address: %p\n", arr2);

    arr1 = (int *)realloc(arr1, 10 * sizeof(int));

    printf("realloc() Address: %p\n", arr1);

    free(arr1);
    free(arr2);

    printf("Memory Freed Successfully\n");

    return 0;
}
```

```
ubuntu@ubuntu:~$ sudo apt install valgrind -y
Installing:
  valgrind

Suggested packages:
  valgrind-mpi  kcachegrind

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 515
  Download size: 9564 kB
  Space needed: 49.1 MB / 1504 MB available

Get:1 http://archive.ubuntu.com/ubuntu/resolute/main arm64 valgrind arm64 1:3.26.0-0ubuntu1 [9564 kB]
Fetched 9564 kB in 4s (2653 kB/s)
Selecting previously unselected package valgrind.
(Reading database ... 215487 files and directories currently installed.)
Preparing to unpack .../valgrind_1%3a3.26.0-0ubuntu1_arm64.deb ...
Unpacking valgrind (1:3.26.0-0ubuntu1) ...
Setting up valgrind (1:3.26.0-0ubuntu1) ...
Processing triggers for man-db (2.13.1-1build1) ...
ubuntu@ubuntu:~$ valgrind --version
valgrind-3.26.0
ubuntu@ubuntu:~$ valgrind ./memory_management
==8095== Memcheck, a memory error detector
==8095== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==8095== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==8095== Command: ./memory_management
==8095==
malloc() Address: 0x4aae040
calloc() Address: 0x4aae4e0
realloc() Address: 0x4aae540
Memory Freed Successfully
==8095==
==8095== HEAP SUMMARY:
==8095==    in use at exit: 0 bytes in 0 blocks
==8095==    total heap usage: 4 allocs, 4 frees, 1,104 bytes allocated
==8095==
==8095== All heap blocks were freed -- no leaks are possible
==8095==
```

```
==8095== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
ubuntu@ubuntu:~$ valgrind --leak-check=full ./memory_management
==8102== Memcheck, a memory error detector
==8102== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==8102== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==8102== Command: ./memory_management
==8102==
malloc() Address: 0x4aae040
calloc() Address: 0x4aae4e0
realloc() Address: 0x4aae540
Memory Freed Successfully
==8102==
==8102== HEAP SUMMARY:
==8102==    in use at exit: 0 bytes in 0 blocks
==8102==    total heap usage: 4 allocs, 4 frees, 1,104 bytes allocated
==8102==
==8102== All heap blocks were freed -- no leaks are possible
==8102==
==8102== For lists of detected and suppressed errors, rerun with: -s
==8102== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
ubuntu@ubuntu:~$ nano cow.c
ubuntu@ubuntu:~$ gcc cow.c -o cow
ubuntu@ubuntu:~$ ./cow
Before fork:
Global Variable Address: 0xc07ad6810010
Global Variable Value : 100
```

```
Child Process
Before Modification: 100
After Modification : 200
Address             : 0xc07ad6810010

Parent Process
Global Variable Value: 100
Address             : 0xc07ad6810010
ubuntu@ubuntu:~$ ps -ef | grep cow
ubuntu  8132  6144 20 08:55 pts/0    00:00:00 grep --color=auto cow
ubuntu@ubuntu:~$ v
v: command not found
ubuntu@ubuntu:~$ cat /proc/PID/maps
cat: /proc/PID/maps: No such file or directory
ubuntu@ubuntu:~$ cat /proc/3456/maps
cat: /proc/3456/maps: No such file or directory
ubuntu@ubuntu:~$ pmap 3456
ubuntu@ubuntu:~$ pmap -x 3456
ubuntu@ubuntu:~$
```

```
ubuntu@ubuntu:~$ cat /proc/PID/maps
cat: /proc/PID/maps: No such file or directory
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```